

Multi-Camera Calibration With Imperfect 3-D Targets for Imaging Biological Systems

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Abstract—High-resolution multiview stereo camera systems provide an opportunity for precise measurement of shapes, deformations, and displacements. These systems are especially useful for instrumentation and measurement of small-scale biological systems. Often, these are so precise that their measurements have greater precision than the achievable manufacturing precision of calibration targets, limiting the accuracy of template-based calibrations. This work extends the calibration of camera systems using imperfect planar calibration targets to 3-D targets, allowing robust calibration of these high-resolution stereo systems. Results are reported for three separate camera systems, designed for human-scale, skin surface, and single-organ imaging. Recovery of target imperfections, to within 20 μm for a 40-mm target, and improved reprojection errors are demonstrated. The proposed calibration of imperfect targets is shown to be more robust to imperfections in the points chosen to constrain the calibration than previous methods. For systems with relatively precise targets, small improvements are shown in calibration results (from 0.91 to 0.59 pixels of Euclidean error) with limited change in measurements from the system. For imprecise targets, the results show notable improvements in reprojection errors (improving from 4.30 to 1.34 pixels and from 5.85 to 0.61 pixels of Euclidean error) and improvements in reconstruction-based measurement, reconstructing an additional 10% of the subject.

Index Terms—Biological tissues, camera calibration, imaging, optimization.

I. INTRODUCTION

A. Motivation

STEREO imaging-based measurement enables the reconstruction of geometry and deformation. This is useful for many applications enabled by noncontact measurement of biological tissues, including animation, surgical guidance, and personalized medical care [1], [2], [3], [4]. Many of these applications require imaging of relatively low-texture organs with significant deformations, such as the breast [5], lung [6], or liver [3]. Generally, systems that measure these biological properties are high-resolution, inward-facing camera rigs that

surround the tissue of interest, like those described for the heart [7] and lungs [6]. Such sensor systems require precise calibration, as these systems have a designed measurement scale on the order of tens-of- μm , and thus, calibration of these systems is sensitive to even small imperfections in the calibration target. Standard 2-D calibration targets work well when every camera can see the target; however, cameras may not be well constrained relative to each other when cameras cannot see the same points on the target simultaneously. With the inward-facing systems typical in biological applications, 3-D calibration targets help to achieve precise calibrations. Rigid 3-D calibration targets can be used to gather closely coupled data for each camera in every frame, but these precise targets can be difficult to fabricate to be within tens of micrometers. Imperfect fabrication introduces error, which can substantially impact the performance of these high-resolution, small field-of-view systems. Some calibration methods can jointly recover the target geometry, accounting for imperfections [8], [9], [10]. However, the implementation and assessment of such models have been limited to 2-D targets [9], [11], [12].

This study describes the implementation of robust calibration of high-precision camera systems using imperfect targets and an efficient solution of the associated bundle-adjustment problem while preserving accurate scale in 3-D targets.

B. Camera Calibration

A camera model [13] describes how a camera, c , projects from world coordinates $\mathcal{W}$ to image domain coordinates, $(u, v) \in \Omega_{c,i} \subset \mathbb{R}^2$, at a given time i . The pinhole camera has ten parameters and is usually paired with the Brown–Conrady model of lens distortion [14], [15], which accounts for radial and tangential distortion. Fisheye cameras are commonly used to represent cameras with very wide fields of view, although more expressive models can represent arbitrary projective behaviors [9].

Camera calibration determines the appropriate parameters for one or more camera models via bundle adjustment [16]. Bundle adjustment allows many parameterizations of these features: by the pose of a known template [17]; fully independent [16], [18]; or a hybrid approach [9], [11], [19]. Template-based calibrations use known patterns to improve detection speed and robustness [8], [20], [21]. The majority of calibration targets are planar, following [17]. However, alternative geometries are advantageous [8], [22], [23], [24] for calibration of some camera systems. This calibration approach

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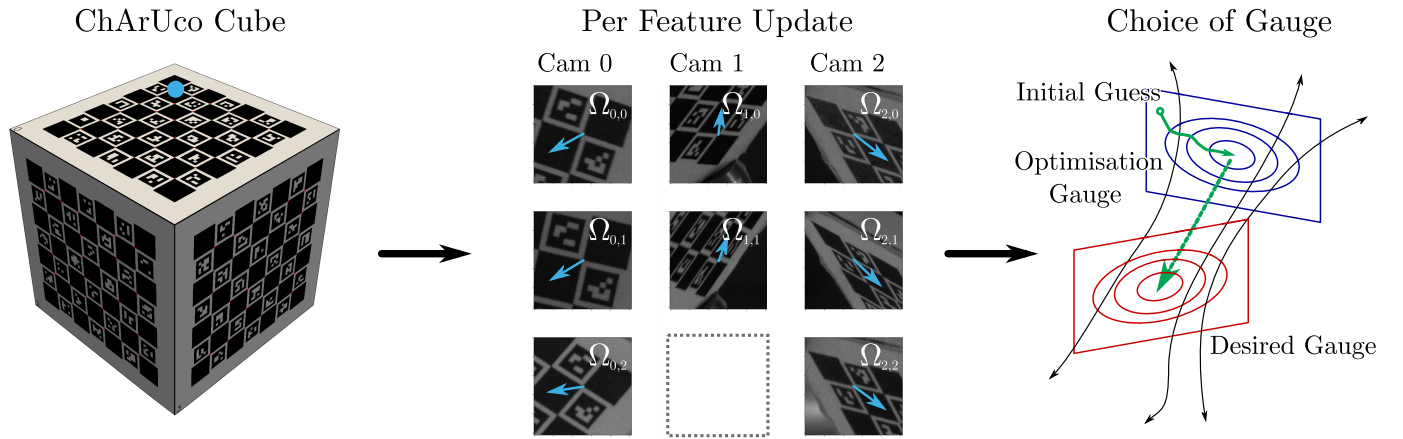


Fig. 1. Overview of the proposed camera calibration methodology. A 3-D calibration target is defined as the combination of unique ChArUco targets, each of which defines a checkerboard of local features. The location of each local feature on the target (e.g., blue dot) is optimized for each pose and across all cameras by adjusting the camera model parameters. This updates the location of the local feature for every image at each pose and camera (e.g., blue arrow). Allowing local feature updates introduces gauge symmetries, continuous and infinite families of parameters with invariant reprojection error. The desired gauge with correct scale, rotation, and translation is indicated in red, and the gauge in which the optimization (green line) is performed is in blue (inspired by [16]).

forms a large nonlinear least-squares problem [13], typically solved via Levenburg–Marquadt iteration [16], [25], [26], [27].

A calibration target used for these template-based calibrations is a collection of features [9], [21] in a local coordinate system $\mathbf{m} \in \mathbb{T}$. For a given time i , the target has pose $\mathbf{T}_i : \mathbb{T} \rightarrow \mathcal{W}$, i.e., a mapping describing the position of the calibration target features in world coordinates, $\mathcal{W}$.

C. Imperfect Calibration Targets

An imperfect target has unknown physical deviations from its computational design, as it can only be constructed with limited precision. Printed targets can typically be constructed with precision of $\pm 50 \mu\text{m}$, as high-resolution printers are readily available [11], but the increased complexity of 3-D targets can result in substantial imperfections, which limit the achievable reprojection error [8]. A computational parameterized calibration target is typically introduced to represent these imperfections. These parameterizations range from modeling explicit forms of error, such as relative rigid body motion between the faces of a target [8], [22], [24] to optimizing each feature $\mathbf{m} \in \mathbb{T}$ [9], [10] to match the manufactured target. For 2-D targets, corrections generally consist of 3-D updates to the target geometry [28]. Using this type of parameterization, it is possible for a target to be scaled, rotated, and/or translated without increasing the reprojection error (known as gauge symmetries [16]), leading to many or infinite solutions for calibration. To avoid multiple solutions, in part, three points of the calibration target can be fixed [29], although this approach overconstrains the target geometry, since errors in these points cannot be represented.

D. Challenges With Camera Calibration

Precise camera calibration is generally data-intensive, typically requiring many features simultaneously visible in multiple cameras [8], [24], [30]. When an imaging system is sufficiently sensitive to capture target imperfections and these are not modeled in the camera calibration strategy, a consistent bias in the estimated parameters manifests [11], [12]. Existing methods have described parameterizations of these imperfections, but they have either only represented a

subset of errors [8] or forced the accuracy of the solution to depend on a small subset of points [29].

E. Aims and Objectives

The primary aim of this study is to enable calibration of high-precision, inward-facing camera systems with imperfect calibration targets. This is addressed by the following:

- 1) proposing a method to represent imperfections commonly seen in 3-D calibration targets;
- 2) identifying imperfections by jointly calibrating target geometry and camera parameters within a bundle adjustment-based framework;
- 3) assessing the calibrations by the following:
 - a) evaluating the improvement in reprojection error compared to a perfect target;
 - b) validating the recovery of target imperfections;
 - c) measuring the improvement in reconstruction performance compared to calibrations that assume a perfect target;
- 4) releasing the calibration algorithm and software.

This article describes methods for creating and implementing the proposed calibration framework (Section II), applying and evaluating this framework using several multi-camera systems (Section III), and discussing the improvements and their implications for the calibration of multi-camera systems (Section IV).

II. METHODS

Camera calibration requires a robust computational representation of manufactured calibration target imperfections. We describe the modification of the bundle-adjustment formulation to enable the representation of these target imperfections. A high-level overview of the formulation is shown in Fig. 1.

A. Target Design and Manufacture

A calibration target is defined as a finite set of features, $\mathbb{F}$, where each feature, $f \in \mathbb{F}$, has a known position in $\mathbf{m}_f \in \mathbb{T}$. A 3-D calibration target is typically, but not necessarily, a combination of multiple 2-D calibration targets. In this work,

the calibration target is defined as a collection of ChArUco boards, with ArUco targets drawn from unique subsets of the dictionaries defined in [21] (see Fig. 1). Three calibration targets were manufactured: 1) a large-scale (400 mm) cuboid composed of polymethyl methacrylate (PMMA) sheets; 2) a medium-scale (50 mm) cuboid paper net-based target; and 3) a small-scale (30 mm) cuboid paper net-based target. ChArUco boards were laser-engraved into the PMMA surfaces for the calibration target 1), and paper printed at 1200 DPI for calibration targets 2) and 3).

For target 1), the boards were faces of a cube, connected by bolts and brackets at the board corners and edge midpoints. For 2) and 3), calibration targets were manufactured from paper attached to an aluminum base and 3-D printed plastic base, respectively. The features were formed by printing on a net (i.e., 2-D polygons on a plane), which folded over the base object to form a 3-D polyhedron. Using a net facilitates relative alignment between calibration features.

B. Camera Model Selection

Here, we present calibration of linear pinhole cameras with Brown–Conrady distortion [16]. Imaging sensors well explained by this model are typical for biological applications, as they have higher spatial resolution than other models, such as fish-eye cameras and are often utilized by analysis tools used to perform downstream measurements [31]. Nevertheless, the methods proposed are appropriate for any calibration defining a parameterized projection operation. The linear projection is described by an intrinsic matrix, $\mathbf{K}_c$, detailing the camera center and focal lengths, and an extrinsic matrix, $\mathbf{E}_c \in SE(3)$, composed of a rotation matrix, $\mathbf{R}_c \in SO(3)$, and translation, $\mathbf{t}_c \in \mathbb{R}^3$, describing position and orientation of the camera relative to a reference frame. The components of $\mathbf{K}_c$ are the focal lengths along each axis, (f_x, f_y) , and principal axis offsets, (p_x, p_y) , with 0 skew (off-axis components of $\mathbf{K}_c$). This allows the definition of a linear projection from homogeneous 3-D space to homogeneous 2-D coordinates, noting the ambiguity in scale, s , which does not affect the normalized form of the homogeneous 2-vector

$$\mathbf{K}_c = \begin{bmatrix} f_x & 0 & p_x & 0 \\ 0 & f_y & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{E}_c = \begin{bmatrix} \mathbf{R}_c & \mathbf{t}_c \\ \mathbf{0} & 1 \end{bmatrix}$$

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}_c \mathbf{E}_c \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}. \quad (1)$$

We define all camera projection matrices as $\mathbf{P}_c \in \mathbb{R}^{3,4} = \mathbf{K}_c \mathbf{E}_c$, associated image domain, $\Omega_{c,i}$ for images at times i . The distortion model for each camera, $\mathbf{d}_c = [k_1, k_2, k_3, p_1, p_2]$, is parameterized as $\mathbf{d}_c \in \mathbb{R}^5$ using the five-parameter Brown–Conrady model [14]

$$\begin{aligned} u^* &= u_c (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) \\ &\quad + [2p_1 u_c v_c + p_2 (r^2 + 2u_c^2)] \\ v^* &= v_c (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) \\ &\quad + [p_1 (r^2 + 2v_c^2) + 2p_2 u_c v_c] \\ \text{for } r^2 &= u_c^2 + v_c^2, \quad u_c = (u - p_x) / f_x, \quad v_c = (v - p_y) / f_y \\ \text{with } u_d &= f_x \times u^* + p_x, \quad v_d = f_y \times v^* + p_y. \end{aligned} \quad (2)$$

C. Two-Step Target Calibration

Calibration optimizes the camera model parameters ($\mathbf{K}_c$, $\mathbf{E}_c$, and lens distortion parameters, $\mathbf{d}_c$) to minimize the distance between the projection of a feature $\mathbf{w} \in \mathcal{W}$ and the associated detection $\hat{\mathbf{x}} \in \Omega_{c,i}$. Here, a two-step calibration process is used: 1) optimization with the assumption that the target matches the predefined geometry (perfect target calibration) and 2) optimization of the geometry concurrently with the camera parameters (imperfect target calibration). For a perfect target, the location, $\mathbf{m}_f$, of every possible feature, $f \in \mathbb{F}$, is known. As bundle adjustment is nonconvex and sensitive to initialization [16], initially assuming a perfect target improves the robustness of convergence.

Since features are not observed in every image, the camera calibration is performed over the visible subset of images that locate a feature f , $\mathbb{V}_f$. Specifically, this subset contains tuples of time points and cameras, (i, c) , where feature f is visible, defined as

$$\mathbb{V}_f = \{(i, c) \mid i, c \in \mathbb{N} \text{ and } v(i, f, c) = 1\}$$

$$v(i, f, c) = \begin{cases} 1, & \text{if } f \text{ is visible in } \Omega_{c,i} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Every tuple, $(i, c) \in \mathbb{V}_f$ has an associated feature location, $\hat{\mathbf{m}}_{i,f,c} \in \Omega_{c,i}$, which indicates the detection of a point, f , for a given image, i , in camera, c .

1) *Perfect Target Calibration*: For this first step, a multi-camera bundle adjustment is performed for a fixed target across all time steps. Single camera calibrations, giving the projection matrices, $\mathbf{P}_c$, and distortion coefficients, $\mathbf{d}_c$, are obtained with the method of Zhang [17], and used to initialize the multi-camera calibration. The parameters for intrinsic, extrinsic, and distortion models, $\hat{\mathbf{G}} = [\mathbf{P}_0, \dots, \mathbf{P}_{C-1}, \mathbf{d}_0, \dots, \mathbf{d}_{C-1}, \mathbf{T}_1, \dots, \mathbf{T}_{I-1}]$, are obtained as

$$\hat{\mathbf{G}} = \underset{\hat{\mathbf{G}}}{\operatorname{argmin}} \sum_{f \in \mathbb{F}} \sum_{(i,c) \in \mathbb{V}_f} \|\hat{\mathbf{m}}_{i,f,c} - d(\mathbf{P}_c \cdot \mathbf{T}_i(\mathbf{m}_f), \mathbf{d}_c)\|^2. \quad (4)$$

The pose of the calibration target in the first frame is set as the identity, $\mathbf{T}_0 = \mathbf{I} \in SE(3)$. As an alternative, a camera could be set to have $\mathbf{E}_c = \mathbf{I}$.

2) *Gauge Fixing*: When relaxing the perfect target assumption, there is a loss of well-posedness in the optimization. Certain changes in the calibration target description can also be represented as changes in the parameters of the camera model for the proposed objective function presented in (4). This results in nonuniqueness and plateaus in the optimization problem. Specifically, the camera parameters $\mathbf{P}_c$ can be modified so that any rotation $\mathbf{R} \in SO(3)$, isotropic scaling, $s \in \mathbb{R}$, and translation $\mathbf{t} \in \mathbb{R}^3$ applied to the calibration target features $\mathbf{m}$ would not alter the objective computed by (4). All such invariants of the objective function are collectively referred to as a gauge [16].

To ensure well-posedness in the problem, we can use gauge-breaking costs [16], [28] (which introduce nontrivial complexity [32]) or create a fixed gauge by fixing parameters. If the camera calibration is solved in a fixed gauge, it is in principle possible to map the obtained parameters to a different point in the gauge with desirable characteristics [16]. In our specific case, fixing three parameters $\mathbf{m}_0$, $\mathbf{m}_1$, and $\mathbf{m}_2$ creates a fixed gauge, since for this fixed gauge, only the trivial mapping

$g(\mathbf{m}) = \mathbf{m}$ is valid. This fixing resolves any invariants in scale or rotation but restricts the optimization of three points (local feature positions) of the imperfect target.

A less restrictive fixed gauge only fixes two parameters $\mathbf{m}_0$ and $\mathbf{m}_1$ and partially fixes a third feature $\mathbf{m}_2$ that does not fall on the line $\overline{\mathbf{m}_0\mathbf{m}_1}$. After optimizing in this gauge, the optimal parameters $\hat{\mathbf{G}}$ are mapped to a desired gauge, with selected criteria for scale, rotation, and translation. The two fixed parameters prevent changes in scale, as $\|\mathbf{m}_0 - \mathbf{m}_1\|$ is constant, and leave only rotations around the axis $\overline{\mathbf{m}_0\mathbf{m}_1}$ as possible rigid transformations. To determine which features of $\mathbf{m}_2$ to fix, consider the analogous subproblem of preventing rotations around $(0,0)$ within a 2-D plane.

Consider a point $\mathbf{u} = (u, v) \in \mathbb{R}^2$. If neither u nor v is fixed, there are infinitely many solutions to the rotational map $\mathbf{R}\mathbf{u} = \mathbf{u}^*$. However, fixing u and applying the constraint $[0, v] \cdot \mathbf{u} = 0$ forces the unknown v to be perpendicular to the initial position $\mathbf{u}$, defining a coordinate frame where $u = 1$, $\mathbf{u} = [1, 0]$. Any rotation of this point must then satisfy $\mathbf{R}[1, 0]^T = [1, v^*]^T$, which is only satisfied by the identity. Allowing $[0, v] \cdot \mathbf{u} \neq 0$ allows for a nontrivial rotation from $[1, v] \rightarrow [1, -v]$.

Analogously, we use this solution to the subproblem to determine the constraints over $\mathbf{m}_2$. Given the two fixed parameters, only one degree of freedom remains within the plane of rotation with axis $\overline{\mathbf{m}_0\mathbf{m}_1}$. An orthogonal constraint is added by defining $\overline{\mathbf{m}_0\mathbf{m}_2}$ such that $\overline{\mathbf{m}_0\mathbf{m}_2} \perp \overline{\mathbf{m}_0\mathbf{m}_1}$, preventing any rotation around the axis $\overline{\mathbf{m}_0\mathbf{m}_1}$ (see Fig. 2). Note that $\mathbf{m}_2$ can be optimized along the line it defines with $\mathbf{m}_0$.

3) *Imperfect Target Calibration*: In the second step, all $\mathbf{m}_f$ values are no longer assumed to be known exactly and instead become extra parameters in the camera calibration. Thus, calibration target imperfections are treated as optimisable feature locations, following [9].

Using the gauge fixing from Section II-C2, the imperfect target calibration optimizes the same functional cost in (4) for an extended set of parameters, including the fixed gauge restriction

$$\mathbf{G} = [\mathbf{P}_0, \dots, \mathbf{P}_{C-1}, \mathbf{d}_0, \dots, \mathbf{d}_{C-1}, \mathbf{T}_1, \dots, \mathbf{T}_{I-1}, \mathbf{m}_{2,1}, \mathbf{m}_{2,2}, \mathbf{m}_3, \dots, \mathbf{m}_{\|\mathbb{F}\|-1}] \quad (5)$$

which is found by minimizing

$$\hat{\mathbf{G}} = \underset{\mathbf{G}}{\operatorname{argmin}} \sum_{f \in \mathbb{F}} \sum_{(i,c) \in \mathbb{V}_f} \|\hat{\mathbf{m}}_{i,f,c} - d(\mathbf{P}_c \cdot \mathbf{T}_i \cdot \mathbf{m}_f, \mathbf{d}_c)\|^2. \quad (6)$$

A calibration in this gauge can be updated to any preferred gauge, featuring different scales, rotations, and translations. This may be required, because if any imperfections occur in the fixed points $\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2$, they distort the scale and rotation of the calibrated system.

To correct for this, a transformation within the gauge is defined. Overall scale is set by the average distance between adjacent features $\mathbf{m}_f \in \mathbb{T}$. The scaling matrix, $\mathbf{S} \in \mathbb{R}^{4,4}$, is defined as

$$\mathbf{S} = \begin{bmatrix} \mathbf{I}_3 & 0 \\ 0 & \frac{1}{s_f} \end{bmatrix}, \quad \text{where} \quad s_f = \frac{1}{\|\mathbb{F}\|} \sum_{f \in \mathbb{F}} \sum_{h \in H_f} \frac{1}{\|H_f\|} \cdot \frac{\|\mathbf{m}_f - \mathbf{m}_h\|}{s} \quad (7)$$

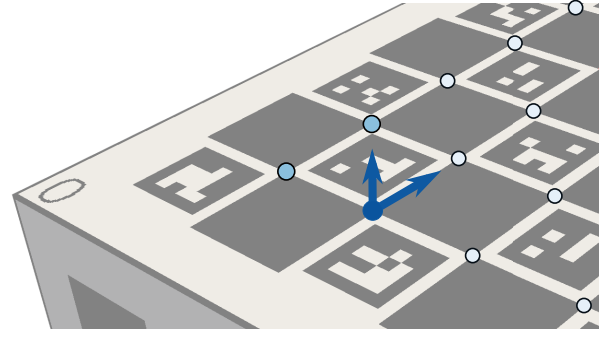


Fig. 2. Example of the proposed feature fixing scheme. One of the 6 ChArUco boards providing texture for the features is visible, with the free features identified as pale blue circles. Fully fixed features are blue circles, and the partially fixed feature is in navy. The degrees of freedom of the partially fixed feature are indicated as navy arrows. Note that one degree of freedom is parallel to the line between the fully fixed points.

where $\mathbf{I}_3$ is the 3-D identity, s is the defined spacing between features, nb is the total number of neighboring pairs, and $H_f = \{h|h \in \mathbb{F}, \|\mathbf{m}_h - \mathbf{m}_f\| = s\}$. Unlike the scale set by the average distance of all feature pairs, this neighborhood scale is not affected by misalignment of the calibration target faces in the 3-D target. However, this neighbor-based scale also benefits a 2-D calibration target, as it is generally robust to compression or bending of the calibration target. While this local approach has benefits in this case, where imperfection-induced scaling is expected due to imperfect base fabrication and alignment, it is not required. A range of scale determination methods (e.g., difference in median scaling) could instead be applied at this step.

A minimal rigid transform $\mathbf{U} \in SE(3)$ is then found on the set of scaled points $\{\mathbf{S}\mathbf{m}|\mathbf{m} \in \mathbf{M}\}$ relative to the original definition using Horn's method [11]. Given the homogeneous scaling matrix, $\mathbf{S}$, and rigid transform, $\mathbf{U}$, which map our parameters $\hat{\mathbf{G}}$ to the desired gauge, we define our final calibration parameters $\mathbf{G}^*$ as

$$\mathbf{E}_c^* = \mathbf{S}\mathbf{E}_c\mathbf{S}^{-1}\mathbf{U}^{-1}, \quad \mathbf{T}_i^* = \mathbf{U}\mathbf{T}_i\mathbf{S}^{-1}\mathbf{U}^{-1} \quad (8)$$

$$\mathbf{m}_f^* = \mathbf{U}\mathbf{S}\mathbf{m}_f^* \quad (9)$$

$$\mathbf{E}_c^*\mathbf{T}_i^*\mathbf{m}_f^* = \mathbf{S}\mathbf{E}_c\mathbf{T}_i\mathbf{m}_f.$$

As projection is scale-independent, the additional scale factor $\mathbf{S}$ present in (9) does not change the results. As a result, the efficient trivial gauge can be used for optimization while maintaining full control over the defined scale.

D. Implementation Details

This calibration method was implemented with the analytical Jacobian of the proposed cost function. While implemented in Python, the cost function/Jacobian was compiled and multithreaded with LLVM using Numba [33]. The calibration was implemented with Python 3.11 and used OpenCV [15] for feature detection, while using SciPy [34] and PETSc-TAO solver routines. The code implemented in this study is available as the PyCamSet library, both on PyPi and on GitHub at <https://github.com/rlav440/pyCamSet>

E. Framework Assessment

To assess the calibration methodology, the reprojection error after calibration, the downstream results of the calibration,

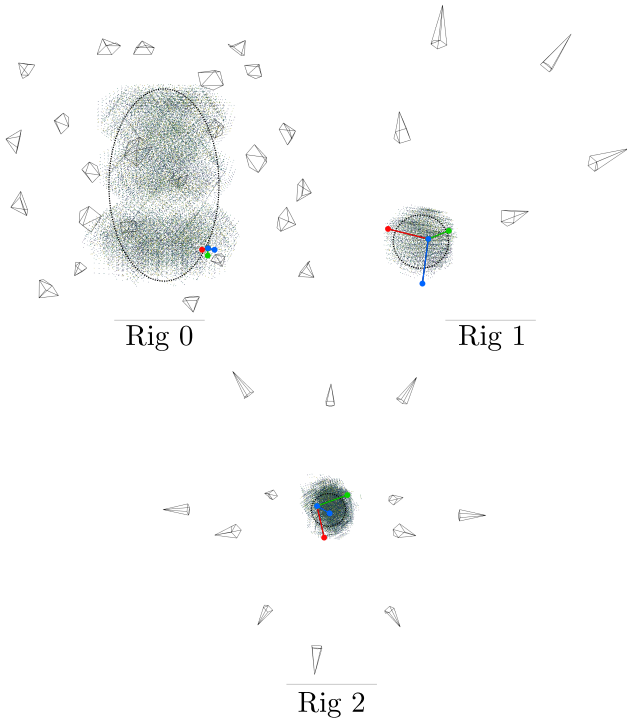


Fig. 3. Per-rig visualization of the relative camera positions, and the data points of the calibration. Each camera, and its field of view, within a rig, is represented by a pyramid. The calibration data fill the imaging region of interest for each rig, which is indicated by a dotted oval.

TABLE I
RELEVANT CAMERA PROPERTIES OF THE THREE RIGS USED FOR
ASSESSMENT OF THE PROPOSED CALIBRATION METHODS

	Cameras	Sensor	Spatial Res.	Target	Subject
Rig 0	24	20 Mp	300 μm	PMMA	Human
Rig 1	5	5 Mp	50 μm	Aluminium	Surface
Rig 2	12	5 Mp	25 μm	Plastic	Organ

and the accuracy of the recovered target geometry were measured. Three datasets were gathered capturing a single set of calibration data on three camera systems: rig 0, a 24-camera human-scale system; rig 1, a 5-camera surface tracking system; and rig 2, a 12-camera small organ imaging system. Rig 0 had 119 images per camera to calibrate, rig 1: 48, and rig 2: 77. Rig 0 aims to capture dynamic whole-body motion, so it uses a surrounding arrangement of cameras. Rig 1 is designed for tracking skin surface texture using five forward-facing cameras with a high spatial resolution. Rig 2 is for whole surface imaging small organs, so it has a small field of view with high spatial resolutions. Each rig was used to evaluate a performance metric relevant to its intended use. These rigs are shown in Fig. 3, and their properties are summarized in Table I.

Rig 0 used the PMMA target, rig 1 used a paper net folded over an aluminum cube, and rig 2 used a paper net folded over a 3-D printed cube. The corner features of both targets were detected using ChArUco board detection and refining algorithms in openCV [35]. As rig 1 had a comparatively narrow field of view, only 3 of the target faces were used for calibration.

For rig 0, the reconstruction performance of the calibrations and the consistency of the arm motion over the range of

interest were assessed. A 125-mm-radius spherical target was mounted on a Universal Robots UR3e (pose repeatability of $\pm 30 \mu\text{m}$) in 30 positions (to sample the working area of rig 0), reconstructed, and then, the position was found in 3-D space. This produced a set of points in the world coordinates, which were rigidly aligned to the reported positions of the UR3e. The residual displacements after alignment were reported as the disagreement-up-to-rigid transformation. The coverage of the reconstruction was evaluated using the 5-mm completeness, calculated as the percentage of the nominal sphere surface within 5 mm of a reconstructed point. As the precise geometry of the target was unknown, a larger completeness threshold was used to account for deviations from sphericity. The RMSE of the reconstructed surface and the completeness adjusted density of the achieved calibration were also reported. Completeness adjusted density was calculated as the number of points on the sphere surface divided by the 5-mm completeness. For rig 1, the ability of the calibration methodology to recover a characterized imperfection was tested. The calibration target used in this system was fabricated to high precision ($\pm 20 \mu\text{m}$), but included a known defect with one surface being intentionally fabricated 1° off-square. For rig 2, the reconstruction performance was evaluated over a small (40 mm) lung phantom. This physical phantom was aligned to both reconstructions using the iterative closest point algorithm (ICP). For this system, the coverage was evaluated with 0.5-mm completeness, and the completeness adjusted density was also reported.

For all calibrated rigs, the performance of the proposed calibration and re-gauging scheme described in Section II-C2 was compared against the calibrations obtained with three fixed points (3FP) [29]. As this approach introduces a strong dependence on the relative fabrication precision of these three points, four further calibrations were performed for each rig. The first set of calibrations performed used favorable fixed points, selected for low update size $\|\mathbf{m}_f^* - \mathbf{m}_f\|$. The second set of calibrations selected adversarial fixed points features around a target corner with expected relative motion $\mathbf{m}_f \rightarrow \mathbf{m}_f^*$.

For each set of fixed points, both 3FP and the proposed method were evaluated. The reprojection errors were computed, and the relative scale, as measured by (7), was detailed for 3FP.

III. RESULTS

A. Recovery of Target Geometries

The recovered geometries are shown in Fig. 4. The target used in rig 0 had a length of 400 mm, and the average required update to the assumed, perfect, geometry was found to be 0.3 mm, with an outlier at 1.3 mm. In general, the updated geometry was smooth, with repeated patterns over the surface.

For the deliberately imperfect calibration target used with rig 1, the joint calibration recovered the angled surface, as illustrated in Fig. 4. The updated surfaces were planar with $15\text{-}\mu\text{m}$ rms error, approximately the nominal accuracy of the fabricated cube ($20 \mu\text{m}$). Of the three visible surfaces, the measured angles between the surfaces were found as 90.01° and 89.98° for perpendicular surfaces, and an angle of 90.93° for the off-square surface. For rig 2, the target had a mean update of 0.37 mm, a planar twist of a calibration target face.

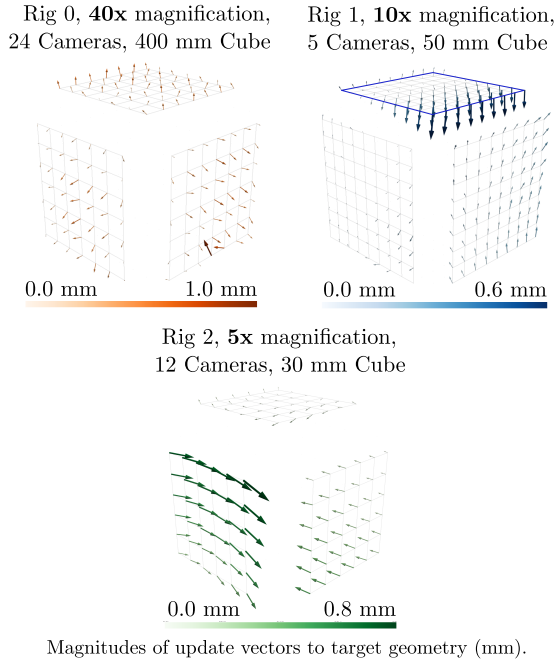


Fig. 4. Representation of the updates to the calibration targets after joint calibration. The PMMA target used with rig 0 displays a small repeated pattern, indicating a minor fabrication defect. The target used for rig 1 clearly shows the recovery of the off-square surface (blue quadrilateral). For the target used with rig 2, the update shows relative ChArUco rotation.

TABLE II
PER-RIG REPROJECTION ERRORS

Method	Rig 0	Rig 1	Rig 2	Mean Time
Template	0.91 px	4.30 px	5.85 px	15.2 s
Joint	0.59 px	1.34 px	0.61 px	24.0 s

TABLE III
CALIBRATION PERFORMANCE WITH FAVORABLE FIXED POINTS

Method	Reprojection Errors	Scale Error
3FP	(0.59 px, 1.34 px, 0.61 px)	(0.1 %, 0.5 %, 0.9 %)
Ours	(0.59 px, 1.34 px, 0.61 px)	-

B. Reprojection Error

The joint calibration improved the calibrations of all rigs, as shown in Table II. The comparatively accurate rig 0 target (Fig. 4) resulted in the smallest improvement.

Calibrating rig 0 showed two error modes: target imperfections and feature mis-detection. Errors larger than 2 pixels were largely conserved between the two methods, shown in Fig. 5. For rigs 1 and 2, the bulk of the initial error was attributable to imperfection, as both had higher spatial resolutions relative to the magnitude of the fabrication errors.

C. Comparison to Three Fixed Points

Comparing against 3FP [29], the results show that fixing favorable points results in equivalent reprojection error and small subpercent scale differences. These results are detailed in Table III. However, if the points have relative fabrication imperfections, the resulting calibrations have worse reprojection error and notable differences in scale, as shown in Table IV. Both methods took approximately the same time (3FP: mean 16.7 s and ours: mean 14.2 s) to solve the bundle adjustment.

TABLE IV
CALIBRATION PERFORMANCE WITH ADVERSARIAL FIXED POINTS

Method	Reprojection Errors	Scale Error
3FP	(0.59 px, 1.42 px, 0.69 px)	(0.2 %, -3.1 %, -5.0 %)
Ours	(0.59 px, 1.34 px, 0.61 px)	-

D. Reconstructions

Over the 30 reconstructions performed with rig 0, there was insubstantial improvement (0.006 mm) in the RMSE error of the sphere fitting and the disagreement-up-to-rigid transformation between the measured target positions from the robot arm (0.03 mm). There were negligible degradations in the relative completeness (0.65%) and density of the reconstructed point cloud using the joint calibration (1% fewer points per sphere). Reconstructing with rig 2 showed a 10% increase in 0.5-mm completeness, from 79.5% to 90.1%. The completeness adjusted density rose from 320 000 to 630 000 points. This corresponds with clear subjective improvements in coverage over the object surface, as seen in Fig. 6.

IV. DISCUSSION

A human-scale imaging system, a surface imaging system, and a single-organ system were calibrated using the proposed novel joint calibration of camera parameters and calibration target geometry. The results demonstrated that both reprojection error and the reconstruction performance benefited from this joint calibration.

Sheet PMMA-based construction of calibration targets produces 3-D calibration targets with only small fabrication defects, while paper nets had characteristically larger errors. In comparing reconstructions from these calibrated systems, the use of a precise target resulted in little difference in the resulting reconstruction and physical metrics. This is consistent with previous literature [9], [12], [28], where small and unbiased errors do not affect the recovered parameters [16]. However, we have shown that when calibration is limited by the geometric precision of the target, a twofold increase in reconstruction density and an additional 10% of the subject could be successfully reconstructed (Fig. 6). This improvement in completeness also much better describes the phantom surface, providing further constraints for downstream measurements, such as those described in [6]. This was despite the construction of rig 2, limiting the possible poses of the calibration target and allowing imperfections to accumulate as bias. The recovered geometry was also shown to precisely match a priori known imperfections (in the calibration of rig 1), rather than low error but arbitrary geometry. Evaluation against 3FP [29] showed increased robustness of reprojection error and increased reliability of the recovered scale when using the proposed gauge fixing and update transformation. For favorable points, there were little to no differences in the reprojection errors, but there were measured changes in the scale of the overall system. This similarity is expected as favorable fixed points effectively fix the system in a trivial gauge, just like the proposed method. If the fixed points were imperfect, the system was barred from properly representing the calibration target. Despite only being $\approx 1\%$ of the object features, these imperfect fixed features decreased calibration performance. There was a large improvement in relative reprojection error, but this came at the cost of inconsistent scaling,

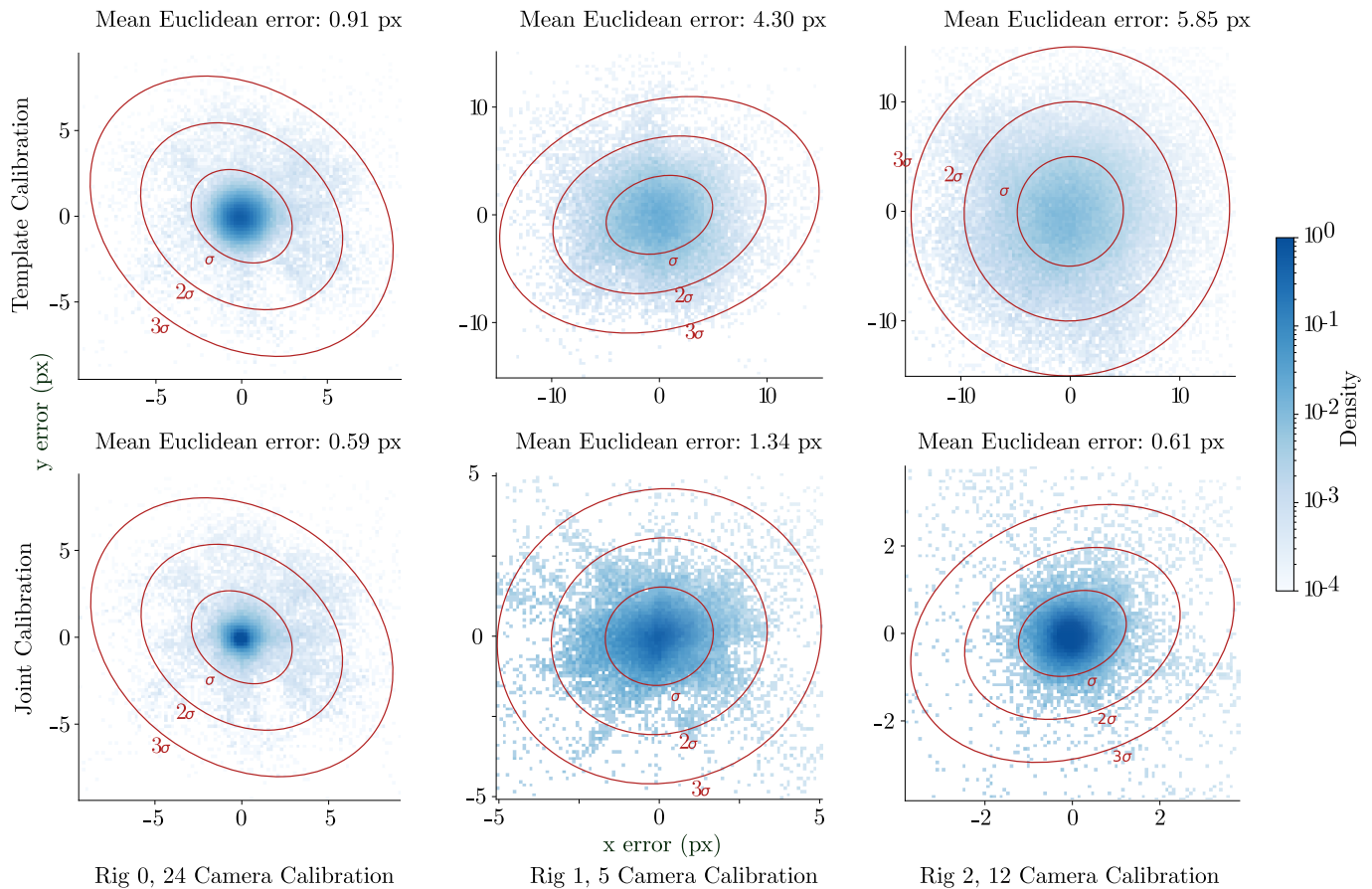


Fig. 5. Histogram plots of the residual errors for the template (top row) and joint (bottom row) calibrations for all camera systems. The use of the joint calibration resulted in a 30% reduction in reprojection error for rig 0 and a 69% reduction in the reprojection error for rig 1 compared to the template calibration. Rig 2 saw a 90% reduction in reprojection error when comparing joint to template calibration. There was little difference in the standard deviation of the residuals for rig 0, despite a visibly tighter central peak of the joint calibration.

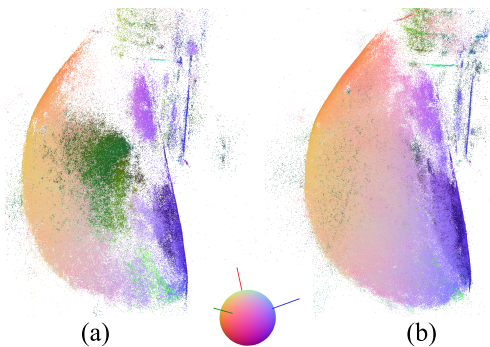


Fig. 6. Comparison of lung phantom reconstructions using multiview stereo reconstruction [31], with a known geometry, using (a) template-based calibration and (b) proposed method. The surface is colored by the normal direction at each point, highlighting shape and completeness. A colored sphere is provided for reference. The template-based calibration has a notably less complete surface and a surface with greater sparsity in reconstructed regions.

which would have notable consequences for comparison of intercalibration measurements. While there are strategies for selecting the fixed points (e.g., features within the same ChArUco board), the absence of local imperfections in the target surface cannot be guaranteed, and these have significant effects on the resulting measurements. After joint calibration, the limiting factor for reprojection error was not the target

geometry, but the detection of corners at oblique angles. This was particularly notable for rig 0, where it is likely that the unchanging variance was the result of misidentified features. This can be observed in Fig. 5, which illustrates little to no observed change in points with initially high error, especially in the calibration of rig 0. Manual inspection confirmed these points as failed detections. While these errors are not ideal, they appear to conform approximately to a Gaussian distribution and thus would likely induce no bias in the final calibration [16]. This is a known issue of 3-D calibration targets and may be addressed by changing target geometries [8] and alternative feature detection methodologies [8], [9], [20]. However, rather than improving the detected feature locations, these methods typically failed to consistently identify features with these systems. The failure occurred as the camera system's nominal spatial resolution began to match the printer resolution (typically around $50 \mu\text{m}$). This demonstrates how high-resolution biological imaging systems can cause issues with otherwise performant algorithms. Despite its age, we found [35] was more reliable in detecting these imperfectly printed features.

V. CONCLUSION

This article has proposed and evaluated a new calibration methodology for stereoscopic imaging systems, with a specific focus on inward-facing multi-camera arrays, which, due to

their high-resolution, typically struggle with defects in 3-D calibration targets. This calibration was achieved by fixing the gauge symmetries introduced by jointly optimizing the calibration target and then mapping to an arbitrary desired gauge. The proposed algorithm obtains high-precision calibrations even when using imperfect targets and was shown to recover the target geometry precisely and improve reconstruction performance in imaging systems when using imprecisely manufactured calibration targets. These systems are particularly common in biological imaging applications, where the improved accuracy allows precise measurement of small and fine-textured biological systems.

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